

SINGLE-USER OPERATING SYSTEMS

UNIT 2. OPERATING SYSTEMS FUNCTIONALITIES AND RESOURCE MANAGEMENT

MICROCOMPUTER SYSTEMS AND NETWORKS

SINGLE-USER OPERATING SYSTEMS

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1. FUNCTIONALITIES OF AN O.S.

The **main functions** of an operating system are:

- **Process** management
- **Memory** management
- **File** management
- **I/O** management
- **Security** management

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2. PROCESS MANAGEMENT

Process management

- A process is a program that is running.
- For a process to begin its execution, it must reside in memory and have all the necessary resources assigned.
- Each process is protected from the rest of the processes, no other will be able to write into the memory areas belonging to that process.
- Each time a program is executed, a process and a data structure called **Process Control Block (PCB)** are created.

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2. PROCESS MANAGEMENT

PCB stores the following information:

- **Process identification (PID):** a unique number assigned by the operating system and different for each process.
- **Process state:** new, ready, running, waiting, terminated
- **Process priority:** it will determine the order in which the processes will run.
- **Assigned memory area:** memory area saved for this process and protected from other processes.
- Process privileges, process structuring information, CPU registers, CPU scheduling information, etc.

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2. PROCESS MANAGEMENT

Task manager

- To open it: right click over the task bar => Task manager.

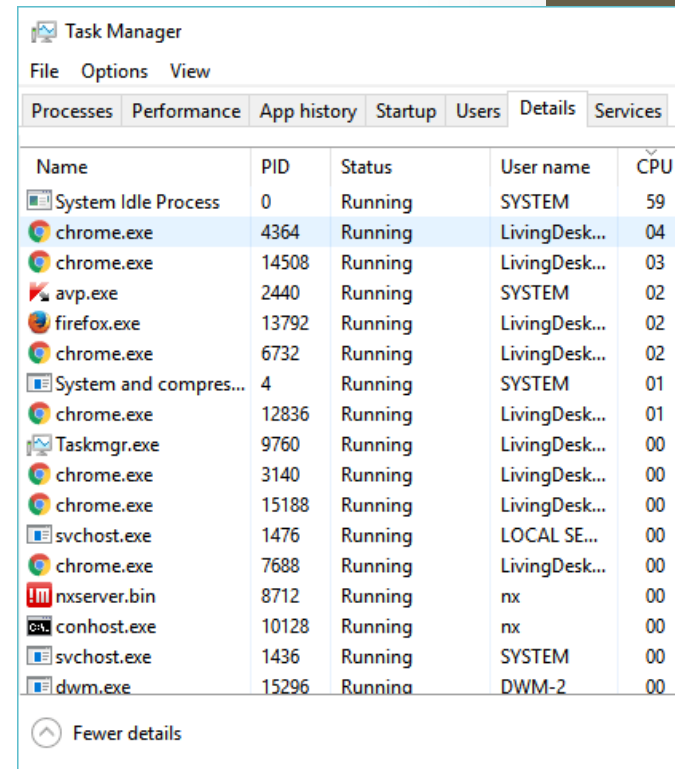
Name	6% CPU	41% Memory	1% Disk	0% Network
Apps (8)				
> Evernote (32 bit)	0.7%	35.1 MB	0.1 MB/s	0 Mbps
> FeedDemon (32 bit)	0%	60.4 MB	0 MB/s	0 Mbps
> Firefox (32 bit)	2.9%	226.6 MB	0 MB/s	0 Mbps
> Internet Explorer (3)	0.3%	167.1 MB	0 MB/s	0 Mbps
> Microsoft Word (32 bit)	0%	30.9 MB	0 MB/s	0 Mbps
> Settings	0%	12.4 MB	0 MB/s	0 Mbps
> Task Manager	0.5%	11.1 MB	0 MB/s	0 Mbps
> Windows Explorer (4)	1.4%	81.0 MB	0.1 MB/s	0 Mbps
Background processes (64)				
> 64-bit Synaptics Pointing Enhanc...	0%	0.6 MB	0 MB/s	0 Mbps
> Adobe® Flash® Player Utility	0%	2.0 MB	0 MB/s	0 Mbps
> Application Frame Host	0%	8.7 MB	0 MB/s	0 Mbps
> Auto Scroll Start Service	0%	0.9 MB	0 MB/s	0 Mbps

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2. PROCESS MANAGEMENT

Depending on who launches a process, we can differentiate **two types** of processes:

- **System processes:** launched by the operating system to provide services to system users. Usually launched at system startup.
- **User processes:** processes that have been **launched by a user** (like Writer or Firefox). We can find processes from two or more users running simultaneously in a MUOS.



The screenshot shows the Windows Task Manager application with the 'Processes' tab selected. It displays a list of running processes with columns for Name, PID, Status, User name, and CPU usage. The processes are sorted by CPU usage in descending order. The list includes system processes like 'System Idle Process' and 'System and compressed memory', as well as user processes like 'chrome.exe', 'avp.exe', 'firefox.exe', 'Taskmgr.exe', 'svchost.exe', 'nxserver.bin', 'conhost.exe', and 'dwm.exe'.

Name	PID	Status	User name	CPU
System Idle Process	0	Running	SYSTEM	59
chrome.exe	4364	Running	LivingDesk...	04
chrome.exe	14508	Running	LivingDesk...	03
avp.exe	2440	Running	SYSTEM	02
firefox.exe	13792	Running	LivingDesk...	02
chrome.exe	6732	Running	LivingDesk...	02
System and compressed...	4	Running	SYSTEM	01
chrome.exe	12836	Running	LivingDesk...	01
Taskmgr.exe	9760	Running	LivingDesk...	00
chrome.exe	3140	Running	LivingDesk...	00
chrome.exe	15188	Running	LivingDesk...	00
svchost.exe	1476	Running	LOCAL SE...	00
chrome.exe	7688	Running	LivingDesk...	00
nxserver.bin	8712	Running	nx	00
conhost.exe	10128	Running	nx	00
svchost.exe	1436	Running	SYSTEM	00
dwm.exe	15296	Running	DWM-2	00

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2. PROCESS MANAGEMENT

System services

- They are **programs** that **run in the background** and provide a certain function to users.
- They are **called background process or daemons** (on Linux) and can be launched by the operating system or by the user.
- In general, these services are used to start, stop, pause, or disable programs or applications on local and remote computers.

2. PROCESS MANAGEMENT

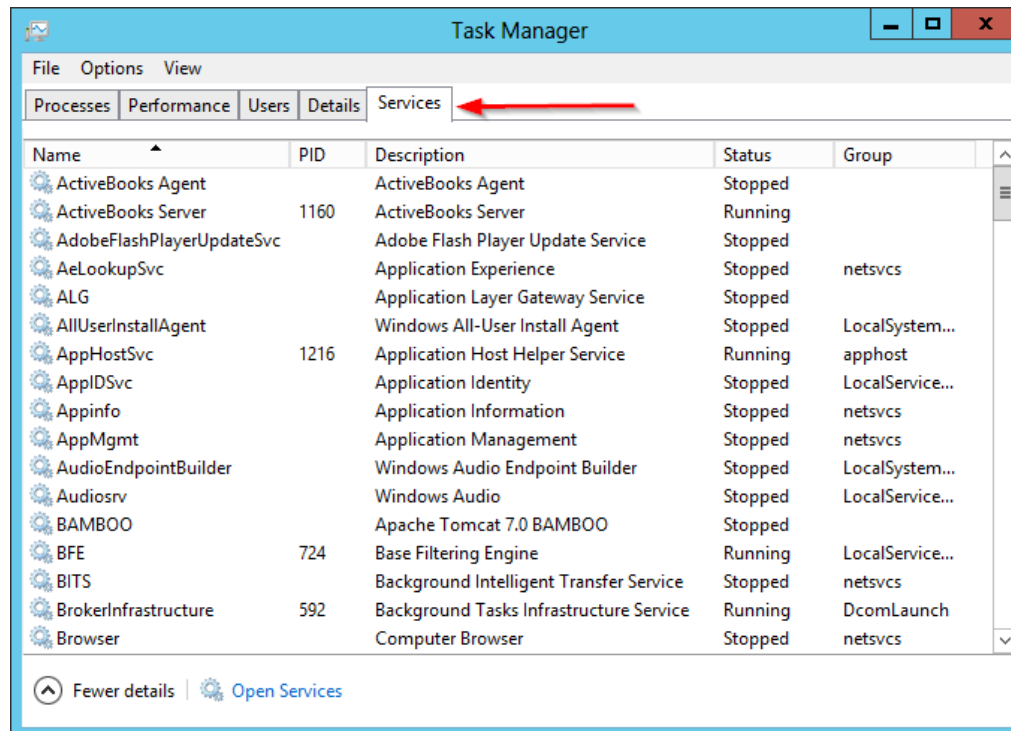
System services

- Some services launched by the OS are client/server style applications, like web servers, database servers, etc.
- For example, the OS needs these services in order to start and stop:
 - ✓ **Start:** prepares the system to be used by the user.
 - ✓ **Stop:** leaves the system in good condition, so it can start up without problems.

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2. PROCESS MANAGEMENT

System services



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2. PROCESS MANAGEMENT

Process scheduling

- The way the CPU is distributed to execute the processes is called **scheduling**.
- **Short-term scheduler**: decides which of the **ready processes** allocated in memory **is to be executed**.

Goals:

- ✓ Minimize response time.
- ✓ Maximize the number of processes executed.
- ✓ Avoid indefinite postponement of processes.

2. PROCESS MANAGEMENT

Process scheduling

- **Medium-term scheduler:** decides which process goes from **main memory** to **secondary memory** or vice versa.
- **Long-term scheduler:** decides which jobs or processes must be **admitted to the ready queue** (in main memory).

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2. PROCESS MANAGEMENT

States of a process

- The states in which a process can be found are:
 - ✓ New
 - ✓ Ready
 - ✓ Blocked
 - ✓ Running
 - ✓ Terminated

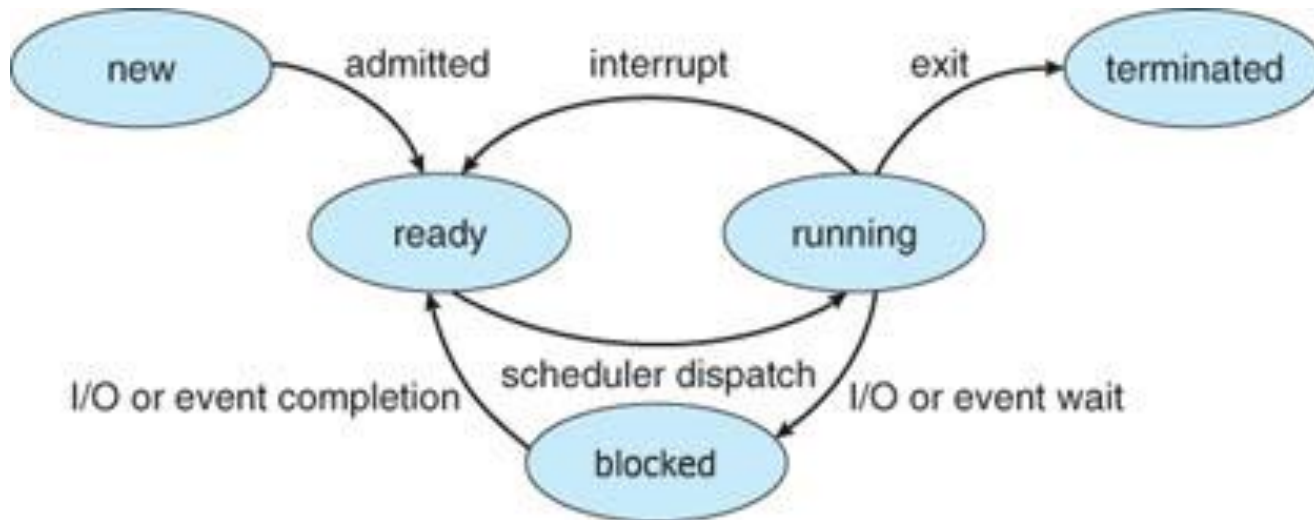


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2. PROCESS MANAGEMENT

State transitions

- **State transition:** occurs when a process goes from a state to another. There are **four possible transitions**:



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2. PROCESS MANAGEMENT

Context switch

- The operation of kicking out a process from the CPU so that another starts running.

Aging

- Technique in which a process with low priority, as long as it waits to use the processor, gains priority in order to avoid being indefinitely postponed due to higher priority processes.

2. PROCESS MANAGEMENT

Scheduling algorithms

- A **scheduling algorithm** is an algorithm used to indicate to the scheduler the **order** in which the processes will be executed, and the execution cycles associated with each one.
- Its **goals** are:
 - ✓ Maximize CPU usage.
 - ✓ Minimize the response time of each process.

2. PROCESS MANAGEMENT

Scheduling algorithms

We distinguish two main types :

- **Expulsive (preemptive):** processes can be evicted from the CPU even if they have not finished their execution and they go to waiting processes queue.
- **Non-expulsive (non-preemptive):** processes cannot be evicted from the CPU until they have finished their execution or they get blocked because they need non-available resources.

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2. PROCESS MANAGEMENT

Scheduling algorithms

- On each algorithm we need to know for every process:
 - ✓ **System arrival time**: the moment when the process enters the system and is ready to run (AT)
 - ✓ **Execution time**: total time the process needs to execute (XT).
- For each process we will be interested in knowing :
 - ✓ **Response time**: total time elapsed between the process reaching the system until results are obtained (RT).
 - ✓ **Waiting time**: the time the process is waiting since it arrives to the system until its execution ends ($WT = RT - XT$)
 - ✓ **Average response and waiting times**: to know how optimal the planning algorithm used is.

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2. PROCESS MANAGEMENT

- **FIFO** (First In First Out). Also called FCFS (First Come First Served)
- **SJF** (Shortest Job First).
- **SRTF** (Shortes Remaining Time First)
- **Priorities Expulsive**
- **Priorities Non-Expulsive**
- **Round Robin**

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2. PROCESS MANAGEMENT

Synchronization and locks

- Several processes may need to share resources such as the screen, the printer or the memory without error situations.
- We can find two types of processes:
 - ✓ **Independent processes:** they do not share resources, so their execution does not affect other processes.
 - ✓ **Cooperating processes:** they share resources, so if at any time one of them need the resource and it is not available, it must wait until it is free to continue executing.

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2. PROCESS MANAGEMENT

Synchronization and locks

- **Mutual exclusion**: mechanism that ensures that only one process will perform a given operation at a time.
- **Critical section**: piece of code of a process that must be executed in mutual exclusion
- **Blocking**: a process will be blocked when it tries to access a resource that is being used by another process in mutual exclusion, so it will not be able to get executed.
- **Deadlock (“abrazo mortal”)**: two processes mutually block each other .

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3. MEMORY MANAGEMENT

Memory management

- Memory manager is the part of the operating system responsible of managing system memory efficiently.
- It will take care of:
 - Keep a **record** of **used and free memory**.
 - **Assign** a memory zone to each process, which will store the **process code**, the **data** and the **stack**.
 - **Control** memory areas **access** and **shared memory**.
 - **Free** the previously assigned memory.
 - Manage the **exchange** between **main memory** and **hard disk**.

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3. MEMORY MANAGEMENT

Memory sharing

- Two processes have a **shared memory area** that they can both access and that will be used to exchange data between them.
- Example: air conditioning and temperature sensor.
- **Memory manager must control access** to these shared memory areas to ensure consistency.

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3. MEMORY MANAGEMENT

Single-tasking system?

- One process after another, sequentially.
- Memory $\Rightarrow$ O.S. + Running process

Multitasking system?

- More than one process running simultaneously.
- Memory $\Rightarrow$ O.S. + Running processes
(partitions)

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3. MEMORY MANAGEMENT

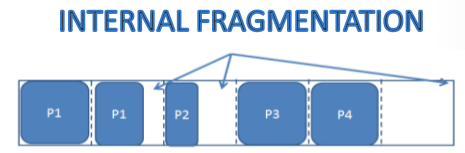
Fragmentation

- Due to memory partitioning.
- Waste of memory.

- Types:

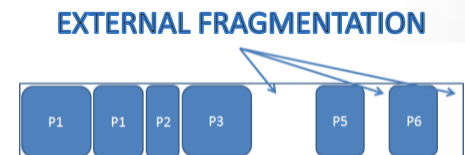
- Internal fragmentation

- ✓ The size of the partition or block is fixed.
 - ✓ Unused partition internal space.



- External fragmentation

- ✓ Variable size of partitions or blocks.
 - ✓ Unused space between different partitions.
 - ✓ Fix: memory compaction.



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3. MEMORY MANAGEMENT

Partitioning techniques

Pagination

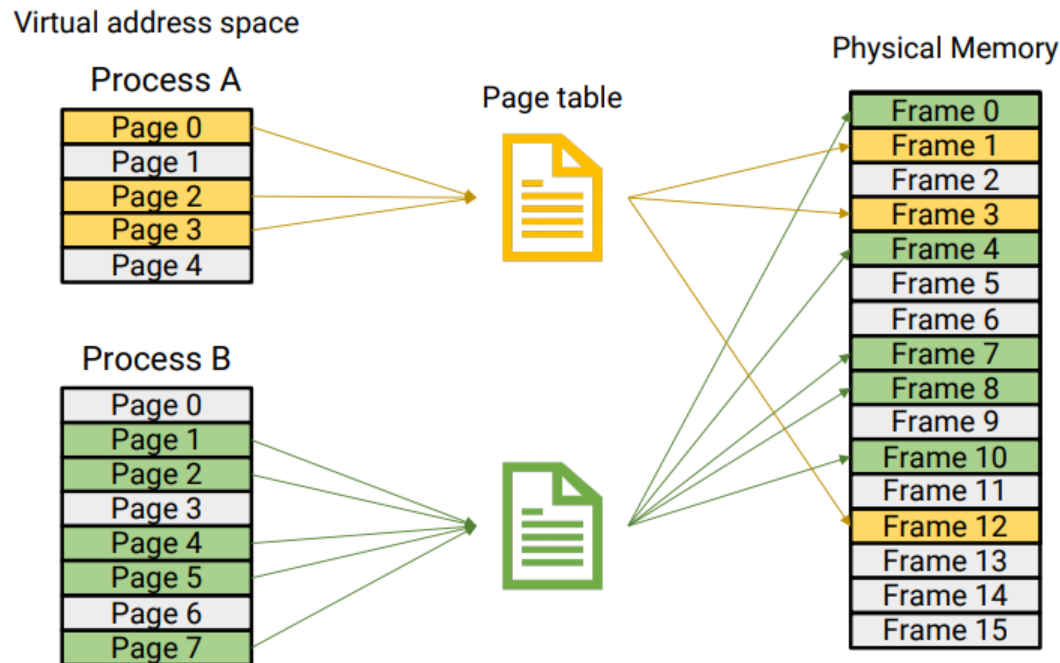
- O.S. divides main memory into fixed-size blocks called **page frames** or, simply, frames.
- The process is divided into **data blocks (pages)** of the same size as the page frames.
- The **pages** that are going **to be used** are loaded into **page frames** in memory.
- A **page fault** occurs when a process requests data from a page that has not been loaded into memory.

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3. MEMORY MANAGEMENT

Partitioning techniques

Pagination



3. MEMORY MANAGEMENT

Partitioning techniques

Pagination

- **Advantages**
 - Avoids external fragmentation.
 - Minimizes internal fragmentation.
- **Disadvantages**
 - Difficulties when running programs larger than the partition.

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3. MEMORY MANAGEMENT

Partitioning techniques

Segmentation

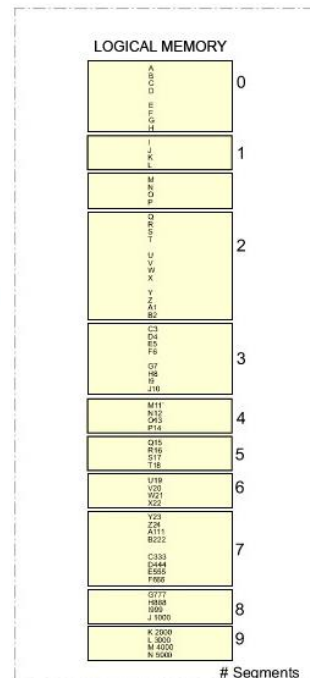
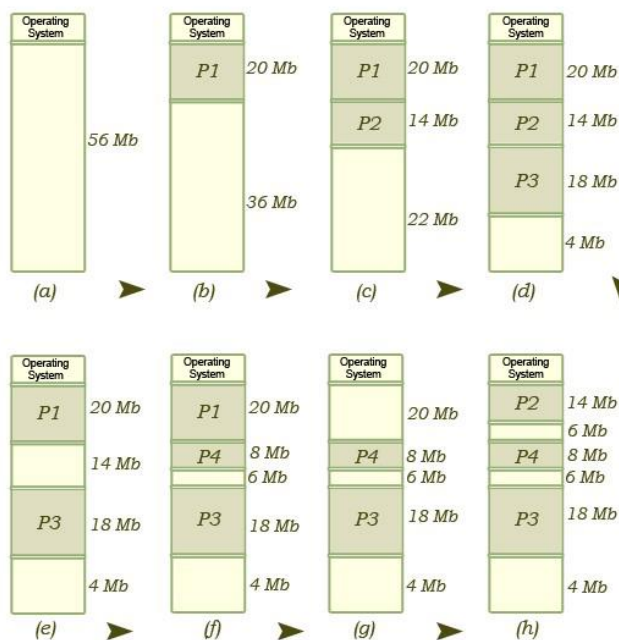
- Memory is divided into zones of **variable size** called **segments**.
- Each process is assigned a segment of the size it needs.
- **External fragmentation** may occur. Fix $\Rightarrow$ **memory compaction** (dynamically relocating processes to optimize memory use)

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3. MEMORY MANAGEMENT

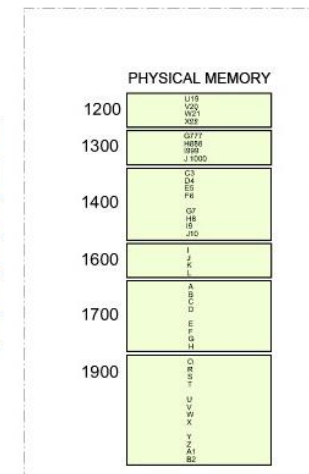
Partitioning techniques

Segmentation



Segment	Physical address
0	1700
1	1600
3	1400
6	1200
8	1300
2	1900

Segments description table



3. MEMORY MANAGEMENT

Partitioning techniques

Segmentation

- ***Advantages***
 - Flexibility
- ***Disadvantages***
 - **External fragmentation** if memory compaction is not done frequently.

3. MEMORY MANAGEMENT

Virtual memory

- Consists in using a space or partition of the hard disk as if it were RAM memory.
 - ✓ Allows us to load more programs into memory.
 - ✓ Allows us to work with larger processes than our real physical memory.
- In Linux: swap partition
- In Windows: C:\pagefile.sys
- When a program is executed, it will be fully loaded into the partition or swap file
- The memory manager will load into main memory only the parts of the process that are required.

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3. MEMORY MANAGEMENT

Virtual memory

- **Pagination and virtual memory**
 1. **Divide** the **process** that will be executed into **pages**.
 2. **Divide** the **swap area** and main **memory** into **page frames**.
 3. **Load all the pages** of the process into the **page frames of the swap area**.
 4. **Load the pages** of the process that need **to be executed** initially in the **page frames of the main memory** assigned to that process.
 5. When data that is not loaded into a memory frame is needed, the **memory manager replaces a page of main memory** with the page of swap memory that is required. This is called a “**page fault**”.

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3. MEMORY MANAGEMENT

Page replacement algorithms

- **FIFO (First Input First Output)**
 - It is the simplest of all. It consists of taking the oldest page from memory.
- **LRU (Less Recently Used)**
 - This algorithm selects the memory page that has not been referenced for the longest time.
 - In current operating systems, improvements to this algorithm are used.

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4. FILE MANAGEMENT

- The file manager is the file system module that takes care of the storage management
- Among the **functions of the file manager** we can find:
 - ✓ Assign data blocks to the files to be stored.
 - ✓ Release the data blocks of a file when it is deleted.
 - ✓ Allow quick access to stored information.
 - ✓ Check that the files do not fill certain disk areas.

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4. FILE MANAGEMENT

- The file managers of current operating systems organize the information through files and folders.
- **File:** Object that represents the logical unit of information storage.
- **Folder or directory:** Special file that contains other files or folders.
- Most current OS allow to organize the file system through directories with a **tree structure**, with a node or source directory called the **root folder**.

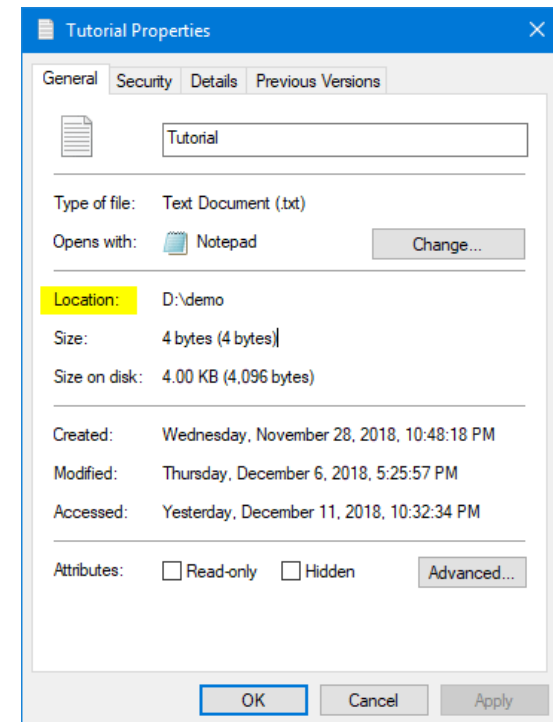
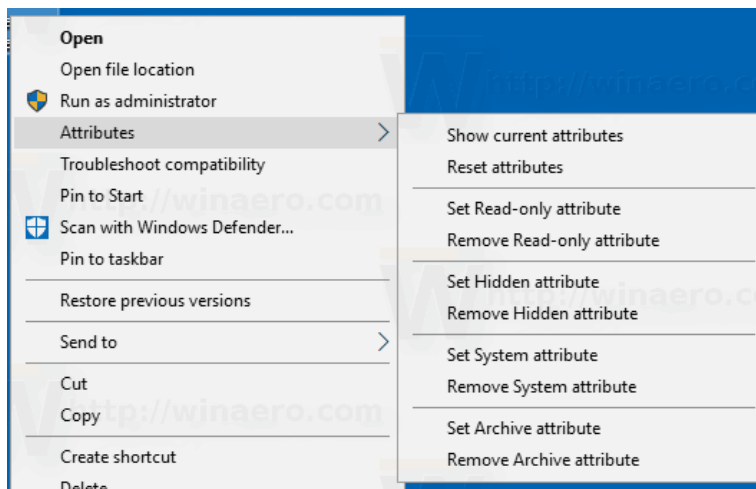


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4. FILE MANAGEMENT

■ Files and folders' properties

- ✓ Name
- ✓ Size
- ✓ Creation/modification/access date
- ✓ Proprietary
- ✓ Permissions
- ✓ Location
- ✓ Attributes

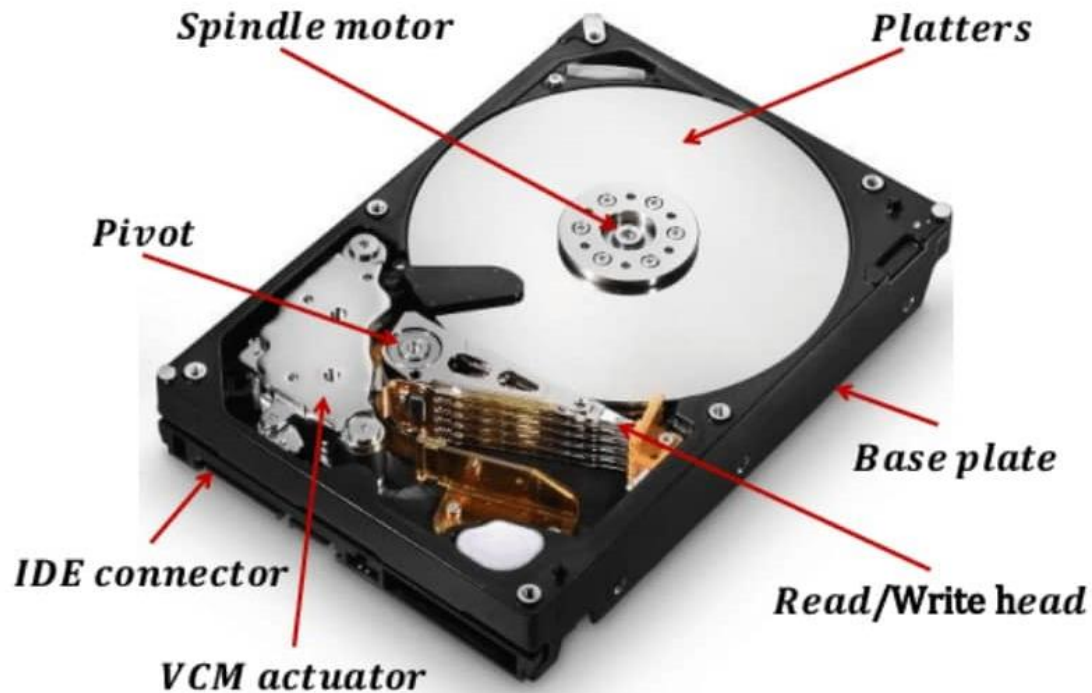


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4. FILE MANAGEMENT

Disk structure

- Physical structure



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4. FILE MANAGEMENT

Disk structure

- Physical structure

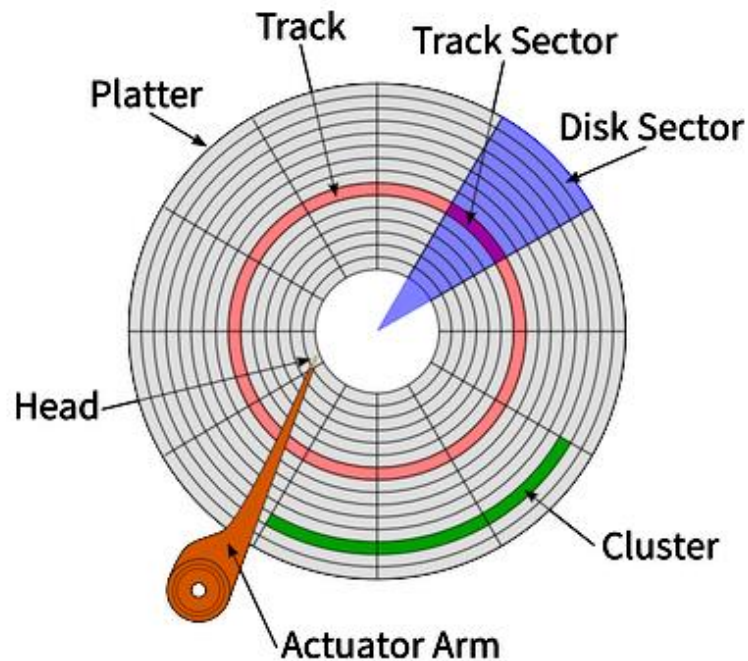


image from <https://medium.com/@andrewjmarkham1>

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4. FILE MANAGEMENT

Disk structure

- Physical structure

- Cluster size?

- Big => Internal fragmentation
- Small => greater delay in reading.
- Linux & Windows: 512 bytes/1KB cluster size
- Configurable in the installation and also when formatting a disk or partition.

Cluster 1 (1KB)		Cluster 2 (1 KB)	
Sector 1 (512 bytes)	Sector 2 (512 bytes)	Sector 3 (512 bytes)	Sector 4 (512 bytes)
Example file (1250 bytes)			Internal fragmentation

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4. FILE MANAGEMENT

Disk partitioning

- Operating systems allow disks to be **divided into multiple logical zones** called **partitions**.
- Each **partition** can work with a **different** type of **file system** (ext4, NTFS, FAT, etc.)
- Structure: Location area & data area
- **Partitioning systems:**
 - ✓ MBR (Master Boot Record)
 - ✓ GPT (GUID Partition Table)

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4. FILE MANAGEMENT

Disk partitioning

- **MBR (Master Boot Record)**

- Standard that began to work in 1983 and that continues to be fully functional nowadays.
- However, as technology advances, this standard is getting obsolete.
- Maximum partition size: 2 TB
- The first physical sector of the disk (zero sector) stores a table with the necessary information to load the operating system.
- BIOS POST process will read the existing table in this sector to launch the OS or boot loader.

4. FILE MANAGEMENT

Disk partitioning

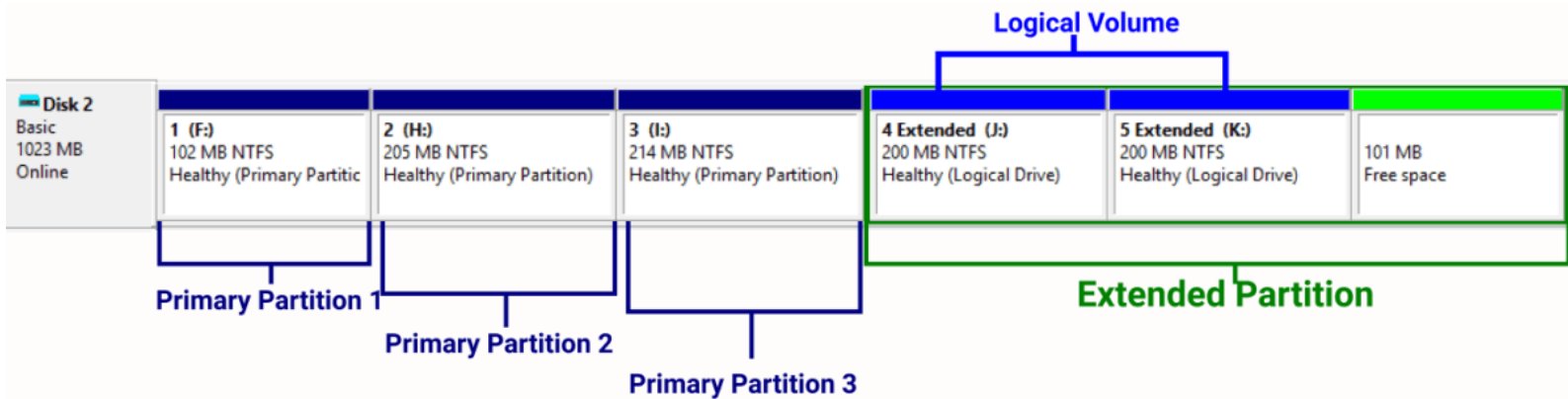
- **MBR (Master Boot Record)**
 - *Types of partitions in MBR:*
 - ✓ **Primary**: up to four on the same disk.
 - ✓ **Extended**: contains logical partitions.
 - ✓ **Logical**: we can have as many as we want, but they must always be located within an extended partition. They allow data storage and were created to remove the limit of four physical partitions.

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4. FILE MANAGEMENT

Disk partitioning

- MBR (Master Boot Record)
- *Partitioning sample*



4. FILE MANAGEMENT

Disk partitioning

- **GPT (GUID Partition Table)**
 - Is the new standard that is replacing MBR.
 - Each partition is associated with a Global Unique Identifier (GUID).
 - No theoretical limit in the number of partitions (Windows has a limit of 128).
 - There only exists primary partitions.
 - Reliability much higher than MBR.

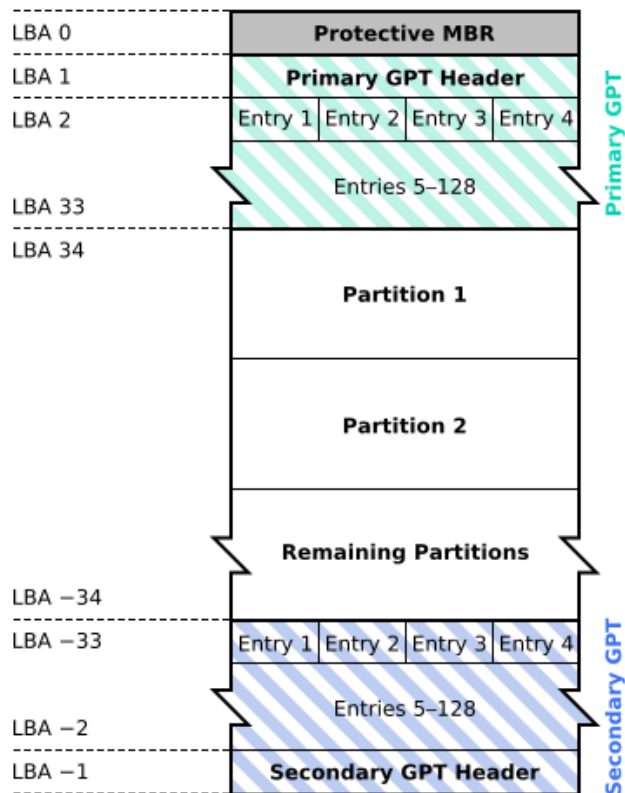
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4. FILE MANAGEMENT

Disk partitioning

- GPT (GUID Partition Table)

GUID Partition Table Scheme



4. FILE MANAGEMENT

Disk partitioning

- GPT (GUID Partition Table)
 - Compatibility:
 - **Windows:** 64-bits versions from Vista onwards are compatibles with GPT.
 - **Linux:** modern versions are compatible.
 - **Apple:** has started to use GPT as the default partition table instead of its own APT (Apple Partition Table).

4. FILE MANAGEMENT

Disk partitioning

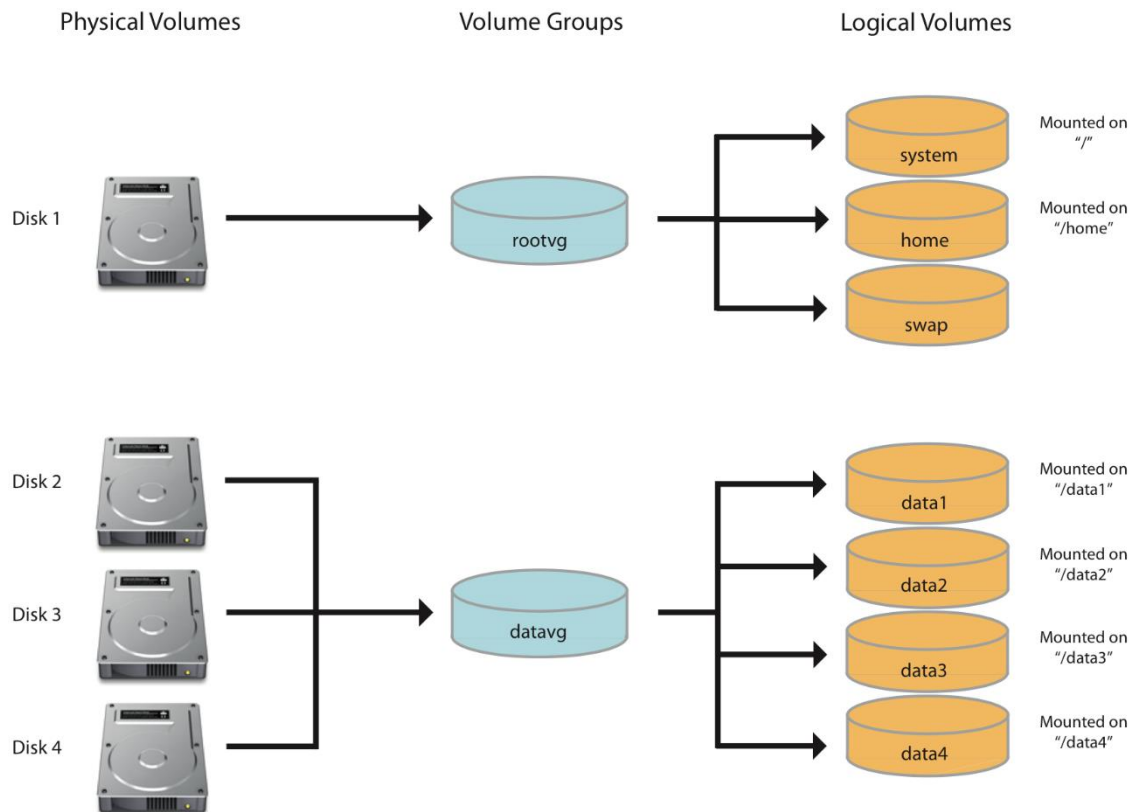
- **Logical Volume Management (LVM)**
 - It allows to create logical volumes made up of several disks or partitions.
 - Each logical volume allows to store information in the same way as in the case of classic partitions.
 - Advantage: if a volume runs out of space, we can expand it without difficulty by connecting another hard drive and adding it to the volume.

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4. FILE MANAGEMENT

Disk partitioning

■ Logical Volume Management (LVM)



4. FILE MANAGEMENT

File system types

- Allows organizing information on disks.
- A different one can be used for each partition
- The file system type is set in the disk formatting process
- Most used: FAT32, NTFS, EXT3, EXT4,...

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4. FILE MANAGEMENT

Most commonly used file system types

■ FAT32 (File Allocation Table)

- ✓ Appeared in the first file managers of Windows systems.
- ✓ It is still in use today since it is often the file system of USB pendrives.
- ✓ **Main characteristics:**
 - The allocation units or clusters are 4KB size.
 - The maximum file size is 4GB (poor).
 - Volumes can go up to 2TB.
 - Access to data is slower than in current systems due to the structure and management of the FAT.

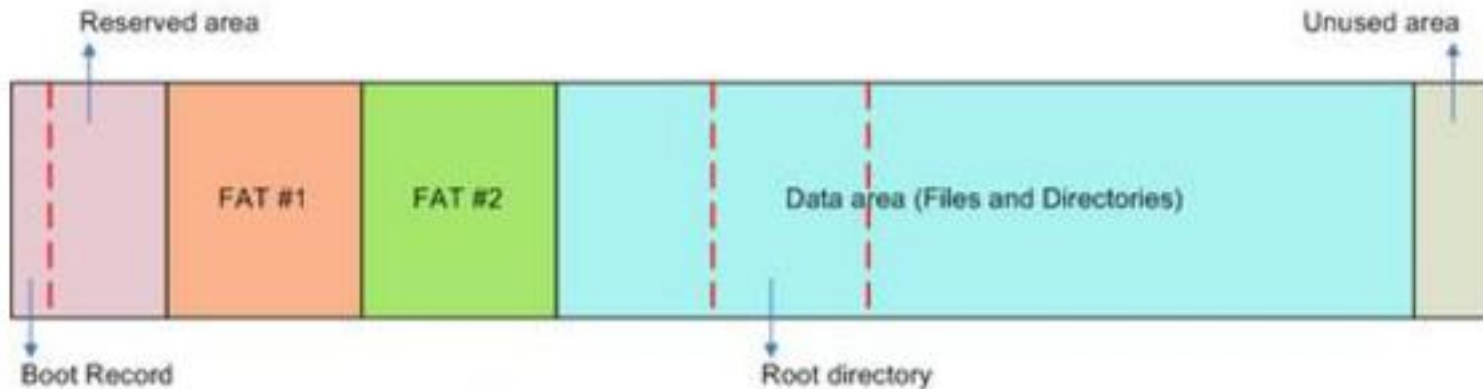
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4. FILE MANAGEMENT

Most commonly used file system types

- **FAT32 (File Allocation Table)**

- ✓ Disk structure in FAT32:



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4. FILE MANAGEMENT

Most commonly used file system types

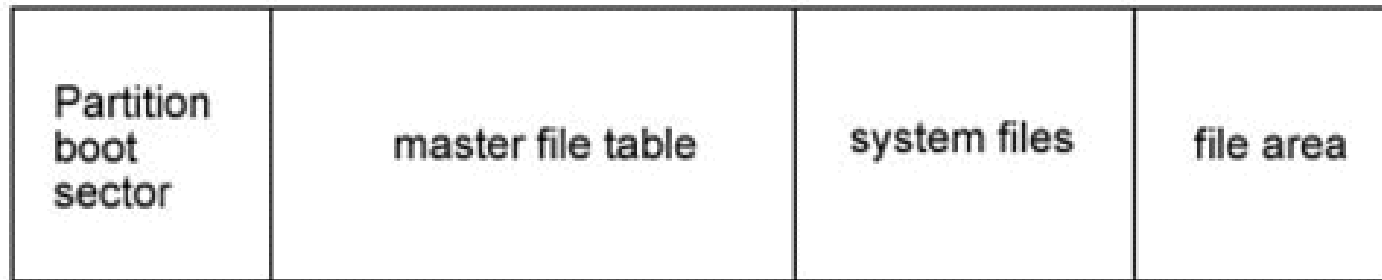
- **NTFS (New Technology File System)**
 - ✓ It is installed by default since Windows 7.
 - ✓ Volumes can go up to 16 TB.
 - ✓ The file size is only limited by the size of the volume.
 - ✓ Allows access attributes on files
 - ✓ Distribution of logical units on several hard drives.
 - ✓ Supports domains, users' disk quotas, etc.

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4. FILE MANAGEMENT

Most commonly used file system types

- **NTFS (New Technology File System)**



4. FILE MANAGEMENT

Most commonly used file system types

- **EXT3 (Extended File System 3)**
 - ✓ Efficient and flexible.
 - ✓ Files are organized by structures called **inodes**.
 - ✓ An inode is a small table associated with each file that contains:
 - Inode number.
 - File's attributes.
 - List with the disk addresses of all its blocks.
 - ✓ When Linux opens a file, the first thing the file manager does is load its inode into memory, so that the process is as fast as possible.

4. FILE MANAGEMENT

Most commonly used file system types

■ EXT3 (Extended File System 3)

✓ Main characteristics:

- The maximum size of a volume is 32 TB.
- The maximum size of a file is 2 TB.
- Is case-sensitive in file and directory names.
- Limited to 32.000 subdirectories.
- **Journaling**: records the changes in the file system to be able to recover the data in case of failure.

4. FILE MANAGEMENT

Most commonly used file system types

- **EXT4 (Extended File System 4)**

- ✓ Main characteristics:

- The maximum size of a volume is 1 EB.
- The maximum size of a file is 16 TB.
- Is case-sensitive in file and directory names.
- Use techniques such as Extents or delayed memory reservation to improve performance and reduce fragmentation.
- Online defragmentation.

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5. INPUT/OUTPUT MANAGEMENT

- Another function of the OS is to manage I/O devices (peripherals and auxiliary memories) and their communication with internal components.
- **Elements involved in I/O:**
 - ✓ I/O manager
 - ✓ Driver
 - ✓ Controller
 - ✓ Peripheral
- **Sample: Software => Printer communication**

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5. I/O MANAGEMENT

Interruption

- It occurs when some hardware element produces a signal to the system to get its attention. They are called IRQs (Interrupt Requests) and they interrupt the work of the processor to assign it to another activity.

DMA (Direct Memory Access)

- It consists of suppressing IRQs and I/O addresses, to transfer information through special lines called DRQ (DMA Request), freeing the processor from workload so the system works faster and more efficiently. Hard drives, graphics and sound cards often use DMA.

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5. I/O MANAGEMENT

Performance improvement techniques

- **Caching**: It consists of storing in a temporary cache for quick access, the data most frequently requested or sent to an I/O device, avoiding the delays produced in the transmission of the information.
- **Buffering**: This technique prevents a slow peripheral from affecting system performance. It consists of using a memory area as a buffer, simulating a device or a logical peripheral, which will act as an intermediate device between the real peripheral and the processor
- **Spooling**: It consists in allowing simultaneous I/O operations to peripherals. Spool is the acronym of “Simultaneous Peripheral Operation On-Line”.

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6. SECURITY MANAGEMENT

- The operating system must control the access of processes or users to system resources. To do this, it must control:
 - ✓ Errors
 - ✓ Incorrect use of the OS
 - ✓ Unauthorized access
 - ✓ Physically
 - ✓ Through the network
 - ✓ By malicious software (viruses, spyware, trojans, etc.)

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6. SECURITY MANAGEMENT

- Although it is impossible for a system to be totally secure, it is important to put the necessary measures to avoid damage to the information or to guarantee its security:

Logical security

- ✓ Use of passwords
- ✓ Encryption of information
- ✓ Firewall activation
- ✓ Use of antivirus software



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6. SECURITY MANAGEMENT

Physical security

- ✓ Use of UPS (Uninterruptible Power Supply),
- ✓ Access control by fingerprint or identification cards, security guards, etc.

